

FraudShield AI

Credit Card Fraud Detection System

Project Report

Algorithm	Random Forest Classifier
Dataset	1,296,675 Credit Card Transactions
Period	2019 – 2020
AUC-ROC	99.51%
Accuracy	98.43%
Recall	93.60%
Tech Stack	Python · Scikit-learn · Streamlit · Power BI · SHAP

1. Executive Summary

FraudShield AI is an end-to-end machine learning system designed to detect credit card fraud in real time. Trained on 1,296,675 real-world transactions, the system uses a Random Forest classifier with 19 engineered features to identify fraudulent activity before it causes financial loss. The model achieves 99.51% AUC-ROC and catches 93.60% of all real fraud cases — meaning out of every 100 genuine fraud transactions, the system flags 93 of them. The project is deployed as a Streamlit web application with live risk scoring, SHAP explainability, and an interactive analytics dashboard.

2. Problem Statement

Credit card fraud costs the global financial industry billions of dollars every year. With millions of transactions happening every second across the world, it is impossible for human analysts to manually review every transaction. Banks and payment processors need automated systems that can flag suspicious transactions in real time — before the money leaves the customer's account.

The core challenge in fraud detection is class imbalance. In real-world data, only a tiny fraction of transactions are fraudulent. In this dataset, just 0.58% of transactions are fraud. A naive model that labels everything as legitimate would achieve 99.42% accuracy — but catch zero fraud. The goal is to build a model that catches as many real fraud cases as possible (high recall) while keeping false alarms at a manageable level.

3. Dataset Overview

The dataset contains real-world credit card transactions from 2019 to 2020, sourced from Kaggle (kartik2112/fraud-detection). It covers transactions across the United States with full cardholder and merchant information.

Attribute	Value
Total transactions	1,296,675
Legitimate	1,289,169 (99.42%)
Fraudulent	7,506 (0.58%)
Original features	23 columns
Engineered features	19 (after preprocessing)
Time period	January 2019 – December 2020
Geography	United States (all 50 states)

4. Feature Engineering

19 features were engineered from the raw transaction data. The original dataset had text columns and datetime fields that needed to be converted into numerical features the model could learn from.

Feature	Type	Description
amt	Original	Transaction amount in USD
hour	Engineered	Hour of transaction (0-23) from timestamp
day	Engineered	Day of month (1-31) from timestamp
month	Engineered	Month (1-12) from timestamp
day_of_week	Engineered	Day of week (0=Monday, 6=Sunday)
age	Engineered	Cardholder age derived from date of birth
distance	Engineered	Euclidean distance between cardholder and merchant
category_enc	Encoded	Label-encoded merchant category
gender_enc	Encoded	Label-encoded gender (M/F)
merchant_enc	Encoded	Label-encoded merchant name
city_enc	Encoded	Label-encoded city name
state_enc	Encoded	Label-encoded state
job_enc	Encoded	Label-encoded cardholder job title
city_pop	Original	Population of the transaction city
zip	Original	ZIP code of transaction
lat / long	Original	Cardholder GPS coordinates
merch_lat/long	Original	Merchant GPS coordinates

5. Methodology

5.1 Handling Class Imbalance

With only 0.58% fraud cases, training directly on raw data would produce a biased model. RandomUnderSampler from the imbalanced-learn library was applied to the training set with a sampling strategy of 0.5, creating a 2:1 ratio of legitimate to fraud transactions. This gave the model sufficient exposure to fraud patterns without discarding all legitimate transaction data.

5.2 Algorithm Selection — Random Forest

Random Forest was selected as the primary algorithm for the following reasons:

- Handles mixed data types (numerical + encoded categoricals) without scaling
- Robust to outliers — fraud amounts vary widely
- Provides native feature importance scores
- Works well with class_weight='balanced' for imbalanced datasets
- Ensemble of 200 trees reduces variance and overfitting
- Fast inference — suitable for real-time scoring

5.3 Model Configuration

Parameter	Value
n_estimators	200 trees
max_depth	20
min_samples_split	5
min_samples_leaf	2
max_features	sqrt
class_weight	balanced
Train/Test Split	80% train / 20% test (stratified)
Random State	42 (reproducible)

6. Model Results

The model was evaluated on a held-out test set of 259,335 transactions never seen during training.

Metric	Score	Interpretation
AUC-ROC	99.51%	Near-perfect separation of fraud from legitimate
Accuracy	98.43%	98.4 out of 100 transactions classified correctly
Recall	93.60%	1,405 of 1,501 real fraud cases caught
Precision	26.09%	Of all flagged transactions, 26% are actual fraud

F1 Score	40.81%	Harmonic mean of precision and recall
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Confusion Matrix — Test Set (259,335 transactions)

	Predicted: Legitimate	Predicted: Fraud
Actual: Legitimate	254,736 (True Negative)	3,098 (False Positive)
Actual: Fraud	96 (False Negative)	1,405 (True Positive)

7. Key EDA Findings

Time Pattern

Fraud spikes sharply between 10 PM and 2 AM, with 22:00 and 23:00 being the highest-risk hours (1,931 and 1,904 cases respectively). Daytime hours between 9 AM and 5 PM show the lowest fraud rates. This suggests fraudsters exploit late-night hours when cardholders are less likely to notice unusual activity.

Category Pattern

grocery_pos (1,743 cases) and shopping_net (1,713 cases) together account for 45% of all fraud. Online merchant categories (suffix _net) generally carry higher risk than in-person categories (suffix _pos), as card-not-present transactions are easier to execute fraudulently.

Amount Pattern

Fraudulent transactions cluster around two ranges: small amounts under Rs.100 (possibly testing if a stolen card works) and a larger range between Rs.600 and Rs.1,200. The average fraud amount of Rs.531 is 7.8 times higher than the average legitimate transaction amount of Rs.68.

Age Pattern

The 46-55 age group has the highest fraud count (1,445 cases), followed by 26-35 (1,417 cases). This suggests middle-aged cardholders are the most common fraud targets, possibly due to higher credit limits.

Geographic Pattern

New York (555), Texas (479), and Pennsylvania (458) are the top three states by fraud volume, which correlates with population density and transaction volume in those states.

8. SHAP Explainability

SHAP (SHapley Additive exPlanations) was applied to explain individual model predictions. For each transaction, SHAP assigns a contribution score to every feature — showing exactly how much each feature pushed the prediction toward fraud or toward legitimate.

Top fraud risk drivers identified by SHAP:

Rank	Feature	Impact
1	amt	Transaction amount — highest single predictor of fraud
2	hour	Time of transaction — late night = high risk
3	distance	Cardholder-merchant distance — large gap = suspicious
4	category_enc	Merchant category — some categories inherently riskier
5	age	Cardholder age — certain age groups more targeted
6	city_pop	City population — fraud more common in dense areas
7	merch_lat/long	Merchant location — unusual locations increase risk

9. Deployment

9.1 Streamlit Web Application

The model is deployed as a Streamlit web application with 5 interactive tabs:

Fraud Checker — Real-time fraud prediction with risk gauge, probability bars, and risk signal explanations

Analytics — Fraud trends by category, hour, amount distribution, and day of week

Model Insights — Confusion matrix, ROC curve, Precision-Recall curve, SHAP plot, and feature importance

EDA — Full exploratory data analysis from 1.29M training transactions

History — Session-level transaction history with risk trend chart

9.2 Analytics Dashboard

An interactive HTML dashboard built with Chart.js provides business-level analytics with 3 dynamic slicers (Category, Gender, Time of Day), 4 KPI cards, and 4 real-time charts. The dashboard updates instantly when filters are applied and can be opened in any browser without any server.

10. Conclusion

FraudShield AI demonstrates a complete production-grade machine learning pipeline — from raw data preprocessing and feature engineering through model training, evaluation, explainability, and deployment. The system achieves state-of-the-art results on a large real-world dataset, with 99.51% AUC-ROC and 93.60% recall — meaning it catches 93 out of every 100 real fraud cases.

The project addresses the key real-world challenges in fraud detection: severe class imbalance, the need for explainable decisions, and the requirement for real-time inference. The combination of a Random Forest classifier, SHAP explainability, and an interactive deployment makes this system suitable for direct use by fraud analysts and business stakeholders.

Future improvements could include real-time data streaming integration, online learning to adapt to evolving fraud patterns, and a more granular geographic analysis using state-level models.